



Envision

Envision BESS Transport Technical Description

Doc number: PMD-ES000006

Date: 2025.09.25

Doc. Title: Envision BESS Transport Technical Description			 Envision
Doc. No.: PMD-ES000006	Rev. No.: A	State:	
			Classification: Internal

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1 Introduction

This document describes the transport requirements and procedures on the Battery Energy Storage System (BESS) of Envision Energy.

2 Safety



Safety is the prerequisite for all the work.

To ensure the safety of operators and BESS equipment, all personnel involved in site work must carefully read and strictly abide by safety specifications. It is the worker's responsibility to meet all safety, local, and general regulations that apply for the specific site, task, or project.

For the specific safety requirements of different processes, refer to the related delivery documentation during contract performance phase.

3 Special Tool List

No.	Description	Specification	Quantity	Unit
1	Container hanging beam	—	1	Set
2	Round sling	15t-9m	4	Set
3	Round sling	25t-4m	6	Set
4	Round sling	20t-5m	2	Set
5	Shackle	20t	4	Set
6	Shackle	17t	4	Set
7	Container hook	CH20	4	Set

4 Equipment Parameters

4.1 DC BANK

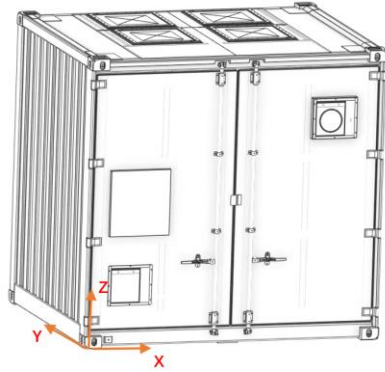
One set of DC system consists of two non-standard 10-foot DC BANK (Battery Segment), one standard 10-foot DC BANK (Combined Segment), and one SKID base, which must be transported separately during shipment. Multiple-layer SKID (for example, 8 layers) can be stacked together for transportation.

Table 4-1 DC BANK parameters

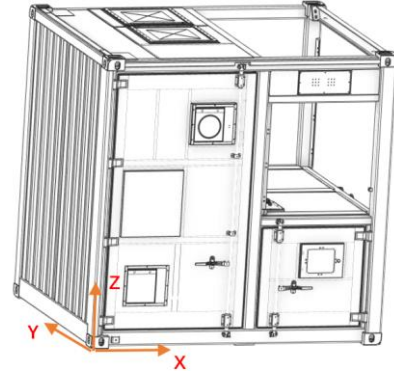
Products	Length (mm)	Width (mm)	Height (mm)	Weight (t, ±3%)	Center of Gravity Position (mm)
DC BANK (Battery Segment)	3091	2438	2896	29	X=1523, Y = 1302, Z =1557

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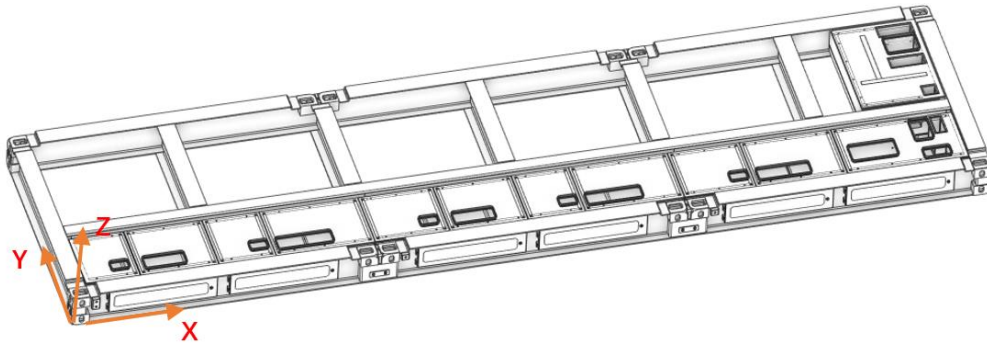
DC BANK (Combined Segment)	2991	2438	2896	18	X=1071, Y =1298, Z =1525
SKID (Single-layer stacking)	9225	2438	360	2.5	X=4629, Y =1052, Z =174
SKID (Multiple-layer stacking)	9225	2438	2896	2.5*N	X= 4629, Y =1052, Z =/



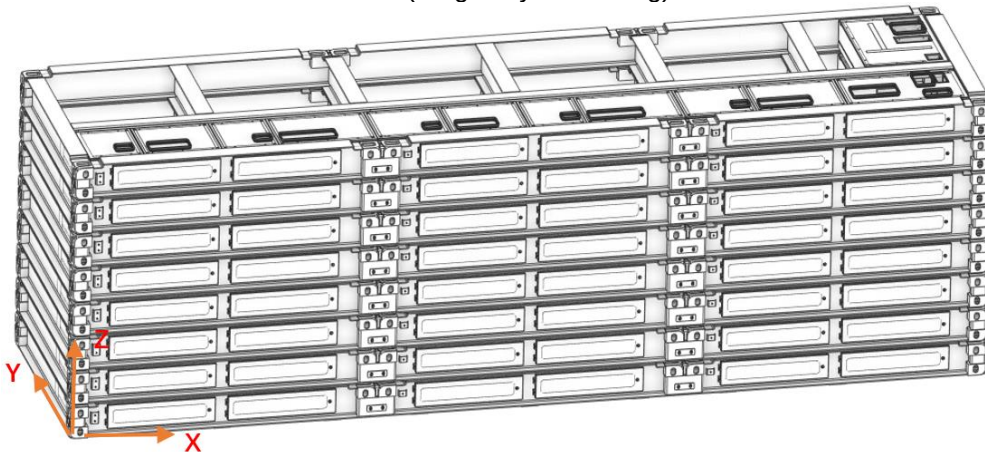
DC BANK (Battery Segment)



DC BANK (Combined Segment)



SKID (Single-layer stacking)



SKID (8-layer stacking as an example)



The figures listed here are for your reference only. The specific details are subject to the actual products.

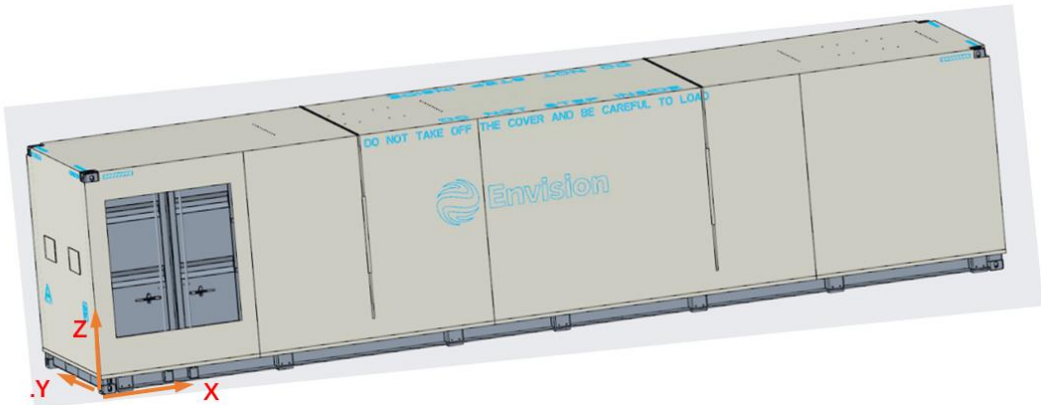
SKID stacking layers are determined based on actual shipping needs

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4.2 AC System

Table 4-2 AC system parameters

Products	Length (mm)	Width (mm)	Height (mm)	Weight (t, $\pm 3\%$)	Center of Gravity Position (mm)
Containerized AC system	12192	2438	2896	49	X=6435, Y = 1219, Z = 1395





The figures listed here are for your reference only. The specific details are subject to the actual products.

5 Transport Requirements

The DC system is of Class 9 hazardous goods, with the liability transferring upon ownership change. Transport regulations for dangerous goods differ by region and shippers must adhere to regional regulations.

5.1 Truck Transport Requirements

Land transportation requires separate low loader trailers or container trucks for the DC and AC equipment.

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No.	Trailer Type	Description	Example
1	Low loader trailer (used for the DC system)	Platform dimensions: - Length: 13.75m/17.5m (The trailer length is for reference only.) - Width: 3 m - Height: 1.5 m/1.2 m	
2	Low loader trailer (used for the AC system with a 40HC-sized exterior)	Platform dimensions: - Length: 17.5 m The trailer length is for reference only. Vehicle load capacity is sufficient. The selected vehicle should ensure the product does not exceed the trailer's loading length. - Width: 3 m - Height: 1.5 m/1.2 m	

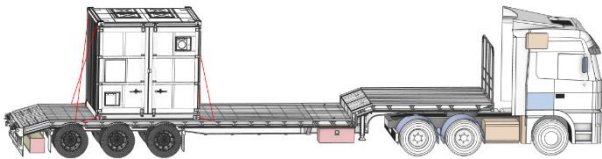
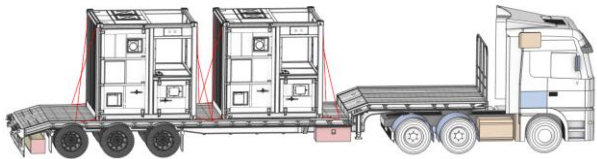
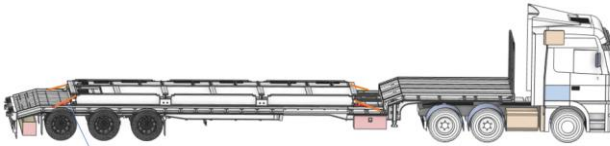
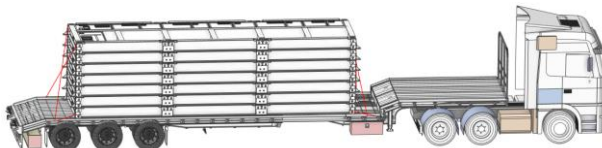
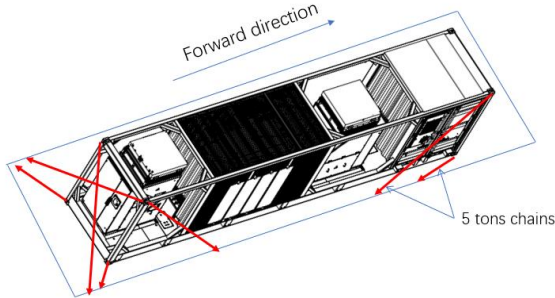


- The selection of road transport vehicles in the destination country must comply with applicable local laws and regulations.
- The vehicle must have a load-bearing capacity sufficient for the cargo weight, a bed width not less than the cargo width, and, where applicable, the required certification for the transportation of hazardous materials in accordance with the cargo type.

5.2 Trailer Loading and Lashing Requirements

The following lashing requirements are the minimum requirements, and the carrier needs to reasonably increase the lashing based on the actual situation.

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No.	Loading and Lashing Requirements	Reference Figure
1	When transported by low loader trailers without positioning cones, 8 chains with a safe load of 5 tons are required for securing the DC BANK, 8-layer SKID and AC system. 4 pieces of 2 tons belts are required for securing the single-layer SKID. At least 4 pieces of 5 mm thick colorless or white rubber padded under each equipment. Do not strap at locations other than the top ISO corner fittings.	 DC (Battery Segment) lashing  DC (Combined Segment) lashing  2 tons belts Single-layer SKID lashing diagram  8-layer SKID lashing diagram  Forward direction 5 tons chains AC lashing diagram
2	When transported by container truck with positioning cones, using their own twist locks for lashing, no longer need lashing chains. The vehicle's load capacity must meet the safety transport requirements. It should be noted that the size of the DC BANK (Battery Segment) is a non-standard 10-foot container.	 

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5.3 Vehicle and Driving Requirements

All vehicles and personnel must comply with the relevant regional laws and regulations. Envision recommends stopping the transportation during extreme weather conditions, for example, torrential rain, dense fog, and so on. The general requirements for transportation vehicles and driver conduct are listed as follows:

- All vehicles must be equipped with no less than three reflective cones, two wood sleepers, and one 3 L effective hand-held fire extinguisher.
- Vehicles must be equipped with GPS configuration during transportation (with positioning and driving speed monitoring functions).
- All vehicle tires must be free of noticeable bulges, exposed steel wires, or sidewall cracks, and must meet the required tread depth for safe operation.
- The maximum vehicle speed must not exceed:
 - 60 km/h on bituminous roads (Abide by the local road transportation laws and regulations of the countries)
 - 40 km/h on cement roads
 - 30 km/h on access roads, road work, and site roads
 - 15 km/h on speed bump and curves
- Transit vehicles must submit a written report on vehicle dynamics in writing every 24 hours and conduct regular inspections of vehicles and cargo (for example, tires, lashing, appearance of cargo, packaging, and so on).
- The minimum height and width of the road must align with the maximum envelope profile of transportation vehicles and cargo to maintain safety margins.
- The drivers must keep the vehicles running smoothly and avoid emergency braking or sudden acceleration.
- Do not remove the packaging during the entire transportation process.

5.4 Road Requirements

The road requirements for transporting the BESS equipment are as follows:

- Subgrade

The straight subgrade width must be at least 4.5 m, with a compaction degree exceeding 0.94. For problematic soil types such as soft soil and collapsible loess, the subgrade must be filled with high-bearing capacity stones. The staggered lanes are required on the infield road, incorporating features such as fork roads and turns.

- Pavement

The straight road surface must be at least 3.5 m wide, composed of mud gravel or mountain gravel, with a thickness of 10 cm. Weak areas might need to receive a base treatment of 20 cm thickness. Road flatness must be maintained within a 20-m span, and the height variation across the road surface must not exceed 150 mm between the highest and lowest points.

- Turning radius

The turning radius is the curvature radius of the center line of the road, and the minimum turning radius is 25 m. The detailed requirements for turning radius and road width are listed below:

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Turning radius of road center line (m)	25	35	40	60	85
Road width (m)	7.5	6.5	6.0	5.5	5.0

- Vertical curve radius

The vertical curve must have a minimum curvature radius of 100 m, meeting the relevant road design requirements for curve length. To accommodate changes in longitudinal slope between sections, the vertical curve must be designed to prevent the underside of the vehicle from hitting the road.

- Slope

Typically, the maximum longitudinal slope must not exceed 8% (4.6 degrees), but in extreme cases, it must not surpass 14% (8 degrees). If towing is required on slopes exceeding 8%, the owner is responsible for arranging towing services. The maximum cross slope must not exceed 3% (2 degrees).

- Bridge

The bridge must meet the equipment transportation requirements.

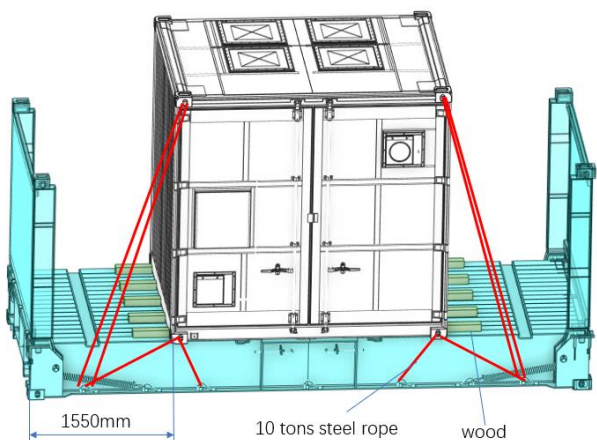
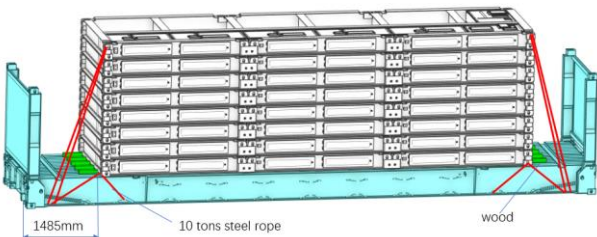
5.5 Shipping Requirements

Bulk carriers are used for shipping, and the lashing methods are standard shipping lashing methods such as twist locks and connecting rod fixings.



If container vessel transportation is adopted, the non-standard 10-foot DC BANK (Battery Segment) and SKID bases shall be loaded into flat rack containers for shipment, while the standard 10-foot DC BANK (Combined Segment) can be coupled in pairs (connection to be completed before ex-works) and shipped as a single 20-foot container. Cargo loaded into flat rack containers must be properly lashed to meet both land and sea transportation requirements.

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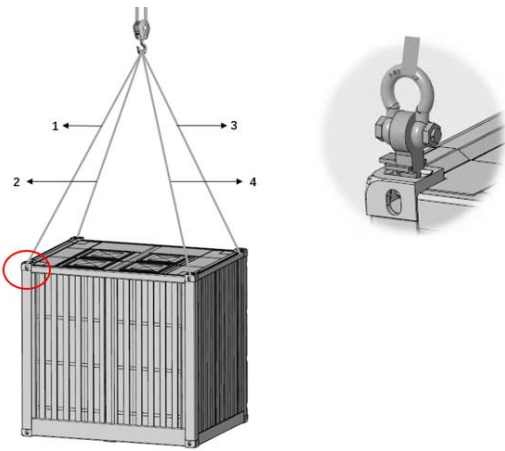
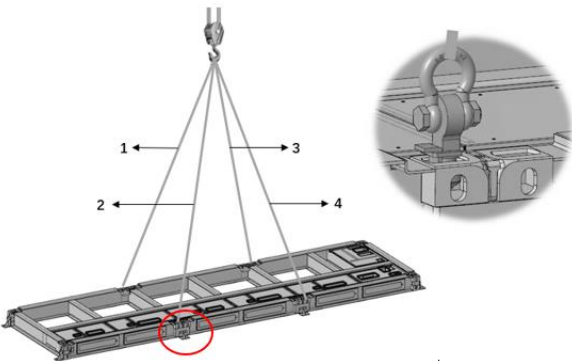
No.	Lashing Requirements in FR	Reference Figure
1	If transported by container ship, the DC BANK and SKID are loaded into a 20/40-foot flat rack (FR) container for transportation. DC BANK and multiple-layer stacking SKID lashing: 10-ton lashing chains or wire ropes for securing, totally 8 lashing points on both sides, the lashing points are the 8 corner fittings on the top and bottom of the cargo. Wooden blocks with no smaller than dimension of no smaller than 400 mm x100 mm x 80 mm must support both ends, total 10 pieces wood. Wooden blocks with no smaller than dimension of no smaller than 400 mm x 100 mm x 80 mm must support both ends, total 4 pieces wood.	 DC load in 20ft FR lashing diagram  Multiple-layer stacking SKID load in 40ft FR lashing diagram

6 Lifting Requirements

5.6 DC BANK Lifting Requirements

Envision recommends using a car crane with lifting tools or a reach stacker for hoisting. Use the lifting beams, slings, and shackles as required. The lifting capacity of the crane needs to meet the requirements.

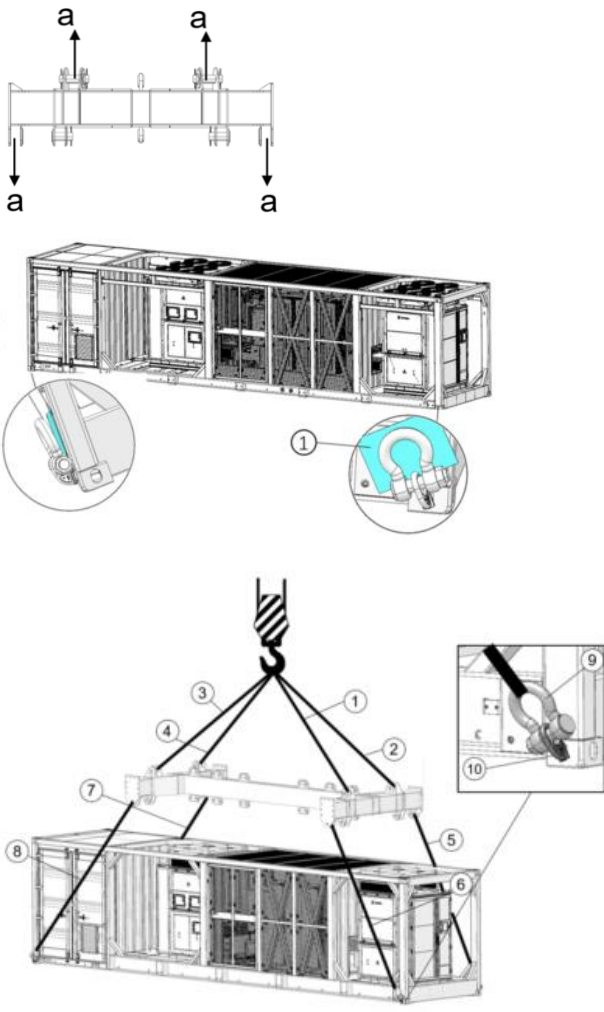
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No.	Lifting Requirements	Reference Figure
1	Install a total of four container hooks at the top corners of the container. Four 15t-9m slings connected to the hooks with two 17t shackles. Tie two taglines at the two lifting points on one side of the long side of the container.	 DC BANK lifting
2	Install a total of four container hooks at the top corners of the container. Four 15t-9m slings connected to the hooks with two 17t shackles. Tie two taglines at the two lifting points on one side of the long side of the container.	 SKID lifting
3	A reach stacker can also be used to lift the container top corner parts at the ports. This depends on whether the reach stacker is suitable for lifting containers of different sizes and weights, which needs to be analyzed based on the actual situation.	

5.7 AC System Lifting Requirements

Envision recommends using a car crane with lifting tools or a reach stacker for lifting. The lifting capacity of the crane needs to meet the requirements.

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No.	Lifting Requirements	Reference Figure
1	Lifting the 40ft AC system container: 1) Install a total of four container hooks at the bottom corners of the container. 2) Install a protective pad between the hook and the product 3) Install four 17t shackles on the four hooks. • Four 25t-4m slings at the upper part of the lifting beam • Two 25t-4m slings (⑤/⑥) and two 25t-5m slings (⑦/⑧) at the lower part of the lifting beam, as shown in the figure. Tie two taglines at the two lifting points on one side of the long side of the container. Note: The lifting configuration provided is for reference only. The actual configuration depends on the specific project.	
2	A reach stacker can also be used to lift the container top corner parts at the ports. Pay attention to the center of gravity to keep the lifting stable.	

7 Equipment Temporary Storage Requirements

The temporary storage requirements for the BESS DC and AC equipment are as follows:

- Ensure that the storage surface is flat with sufficient load-bearing capacity.
- Avoid placing containers near high-temperature heat sources.

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- Choose storage areas free from flooding risks and take fire prevention measures.
- Regularly inspect the containers and promptly report any issues to the personnel of Envision Energy. Prohibit any unauthorized repairs.
- In the transport process, if any cargo damage is found, it must be reported promptly. Any unauthorized repair actions are strictly prohibited.

8 Project Site Unloading Requirements

Upon arrival at the project site, refer to the lifting operation method for unloading after release the lashing between cargo and trailer.